LAB EXERCISE 2 (SECJ1023)

PROGRAMMING TECHNIQUE II

SECTION 03, 04 & 06, SEM 2, 2025/2026

PROBLEM SOLVING

1. Write a class named **`Cone`** that finds the **`baseArea`** and **`volume`** of a conical structure. The class' private data is **`radius`** and **`height`**.
   a. Create four class functions that performs the following actions:
      i. Read the two private data.
      ii. Calculate the value for **`baseArea`**, according to the following equation:

      `baseArea = PI * radius * radius`

      iii. Calculate the value for **`volume`**, based on the following equation:

      `volume = (1/3) * baseArea * height`

      iv. Display the two private data, **`baseArea`**, as well as **`volume`**.
   b. Write the **`main()`** function that performs the following tasks:
      i. Declare an object from **`Cone`** class, called **`cn1`**.
      ii. Call the four functions using the sequence in a(i) to a(iv) above.
   c. Change the **`Cone`** class declaration and the member functions' definition to **`Cone`** class with inline member functions.

Sample Output:

```
Enter radius: 7.25
Enter height: 5.3

2 private data:
        Radius = 7.25
        Height = 5.3

Output:
        Base area = 165.151
        Volume = 291.767
```

2. Write a C++ program using an Object-Oriented Programming (OOP) approach. Write the program in a single source code file. Do the following tasks:
   a. Write a class named **`Stock`** that contains the following:
      i. A **`string`** data field named **`code`** for the stock's code.
      ii. A **`string`** data field named **`name`** for the stock's name.
      iii. A **`double`** data field named **`closePrice`** that stores the stock price for the previous trading day.
      iv. A **`double`** data field named **`currentPrice`** that stores the current stock price.
      v. The accessor functions for **`code`** and **`name`**.

vi. The mutator functions for all data fields.
vii. A function named **`changePercent()`** that returns the percentage changed from **`closePrice`** to **`currentPrice`**.
viii. A function named **`display()`** should display **`closePrice`**, **`currentPrice`** and the price change percentage.

b. Write a **`main()`** function that do the following tasks:
   i. Create two **`Stock`** objects, then use an appropriate function(s) to assign values to all member variables.
   ii. Set a new current price.
   iii. Display the price change percentage.

Note: Please use all the member functions defined in the class declaration.

An example of program output:

```
Stock: Maybank (MBB)
Close price  : RM5500.00
Current price: RM5250.00
Price change percentage: -4.55%

Stock: CIMB Bank Berhad (CIMB)
Close price  : RM10000.00
Current price: RM15050.00
Price change percentage: +50.50%
```

3. Write a C++ program using an Object-Oriented Programming (OOP) approach. Write the program in a single source code file. Do the following tasks:

a. Write a class named **`Painting`** that has the following member variables:
   i. **`artist`**: A string that holds the painter's name who draws the painting.
   ii. **`yearPainted`**: An **`int`** that holds the year the painting was drawn.
   iii. **`auctionPrice`**: A **`double`** that holds the painting's auction price.

   In addition, the class have the following member functions:
   i. Appropriate mutator functions to store values to an object's **`artist`**, **`yearPainted`** and **`auctionPrice`** member variables.
   ii. Appropriate accessor functions to get the values stored in an object's **`artist`**, **`yearPainted`** and **`auctionPrice`** member variables.
   iii. The **auction** function should add **500** to the **auctionPrice** member variable each time it is called.
   iv. The **`displayPrice`** function should display the **`auctionPrice`** value each time it is called.

b. Write the **`main()`** function that does the following tasks:

   i. Creates a **`Painting`** object then, call an appropriate function(s) to assign the value to all member variables.
   ii. Calls the **`auction`** function five times.
   iii. After each call to the **`auction`** function, get the current **`auctionPrice`** of the

painting and display it.

An example of program output:

```
Artist name: Abraham
Year painted: 1975
Starting auction price: RM 14500.00
Current auction price = RM 15000.00
Current auction price = RM 15500.00
Current auction price = RM 16000.00
Current auction price = RM 16500.00
Current auction price = RM 17000.00
```

4. UML diagram for class **`Chemical`** is given in Figure 1. Based on the diagram, declare the class with the given attributes and methods. Create a driver application to test all the members defined by instantiating various objects.

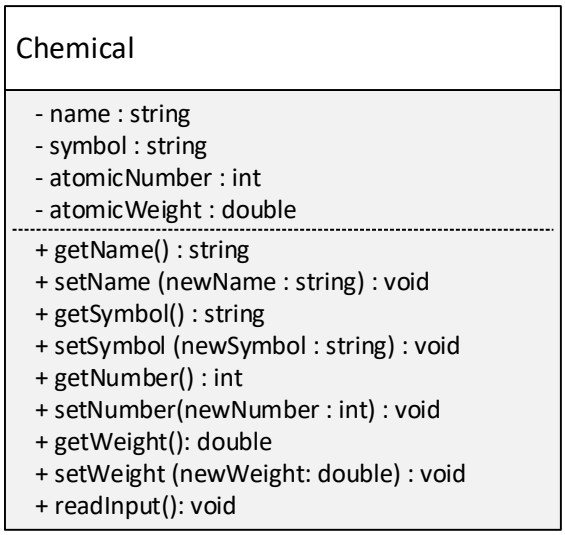

Figure 1: UML class diagram

Sample Output:

```
Please enter the requested information to create a new chemical
object.
Enter the chemical's name: Hydrogen
Enter the chemical's symbol: H
Enter the chemical's atomic number: 1
Enter the chemical's atomic weight: 1.00794
You just entered information for Hydrogen.

And here are its details: Hydrogen [H, #1, 1.00794]
Here is another chemical: Sodium [Na, #11; 22.98977]
```